RESEARCH ARTICLE

Impact of Antidepressant Use and Serotonergic Profile on Short-Term Outcome of Traumatic Brain Injury

Retrospective Nationwide Cohort Study

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Neurology® 2026;106:e214602. doi:10.1212/WNL.0000000000214602

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Abstract

Background and Objectives

Concerns have been raised that the use of serotonergic antidepressants may increase the risk of intracranial hemorrhage and worsen outcomes following traumatic brain injury (TBI). However, large-scale evidence on their impact on TBI outcomes remains limited. Our aim was to investigate the association between preinjury antidepressant use, antidepressant type and serotonergic profile, and the short-term outcome of TBI, focusing on mortality, acute neurosurgical operations (ANOs), and length of hospitalization.

Methods

This nationwide, retrospective cohort study included all patients aged ≥16 years admitted with TBI (ICD-10 S06.*) in Finland between 2005 and 2018. Preinjury antidepressant (Anatomical Therapeutic Chemical codes N06A* and N06CA*) use was identified from national prescription records using daily pill counting method and categorized by serotonergic profile (weak, intermediate, strong). The primary outcome was 30-day mortality, while secondary outcomes included ANOs and length of hospitalization. Cox proportional hazards models were used for mortality analysis, modified Poisson regression models for ANOs, and linear regression models for hospitalization, adjusting for age, sex, comorbidities, vitamin K antagonists (VKAs) use, admission location, and study year.

Results

Of 54,876 patients with TBI, 7,845 (14.3%) were taking antidepressants at the time of injury. Adjusted models showed no significant association between antidepressant use and 30-day mortality (adjusted hazard ratio 0.98; 95% CI 0.90–1.07; $p = 0.696$). Serotonergic profile or type of antidepressant was not associated with mortality. Antidepressant users had a lower likelihood of ANOs (adjusted relative risk 0.89; 95% CI 0.82–0.97; $p = 0.007$). Length of hospitalization did not differ between groups. Antidepressant use, profile, or type did not modify the association of VKA use with increased mortality or ANOs in the interaction analyses.

Discussion

Preinjury antidepressant use, regardless of antidepressant type or serotonergic profile, was not associated with increased mortality, increased need for ANOs, or prolonged hospitalization in TBI patients. No interaction was found between VKA use and antidepressant use, serotonergic class, or type of antidepressant, suggesting that antidepressants do not exacerbate the adverse impact of VKA on these outcomes. Overall, our results suggest that preinjury antidepressant use does not worsen early clinical outcomes after TBI.

Introduction

Incidental falls are the most common cause of traumatic brain injuries (TBIs) in the aging Western countries.[1] TBIs can be complicated by traumatic intracranial hemorrhage, risk factors

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Glossary

ANO = acute neurosurgical operation; **ATC** = Anatomical Therapeutic Chemical; **DOAC** = direct oral anticoagulant; **HR** = hazard ratio; **ICD-10** = International Classification of Diseases, 10th Revision; **IQR** = interquartile range; **RR** = relative risk; **SNRI** = serotonin-norepinephrine reuptake inhibitor; **SSRI** = selective serotonin reuptake inhibitor; **TBI** = traumatic brain injury; **VKA** = vitamin K antagonist.

for which include age, coagulopathy, and antithrombotic medication use.[2,3] Antidepressants, particularly selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs), are among the most prescribed CNS-acting drugs worldwide.[4] The use of these antidepressants in patients at risk of head injury has raised concerns due to their effects on serotonin-mediated platelet aggregation,[5-7] which can theoretically increase the risk of intracranial hemorrhage.[8]

Previous studies have reported an association between SSRI use and bleeding risk in various populations.[9] There are also studies reporting an increased risk of intracranial hemorrhage and ischemic stroke with serotonergic medications,[10,11] but conflicting results have also been reported.[12] Finnish prospective[13] and retrospective single-center studies[14] have recently reported that serotonergic antidepressants were not associated with an increased risk of traumatic intracranial hemorrhage or adverse events after mild TBI. By contrast, vitamin K antagonists (VKAs), but not direct oral anticoagulants (DOACs) and adenosine diphosphate inhibitors, have been shown to be risk factors for acute neurosurgical operations (ANOs) and mortality after TBI.[15,16] The safety of serotonergic antidepressants—and their possible modulating influence with VKA—needs further research in this context, especially regarding broader outcomes such as short-term mortality, length of hospital stays, and the need for ANOs.

We have previously reported an association between preinjury use of CNS-acting medications and worse short-term outcomes after TBI. Preinjury use of various CNS medications including antidepressants was more common in patients with TBI than in matched control groups and was associated with higher short-term mortality.[17] In addition, the use of drugs affecting the CNS was found to be prevalent in patients who died due to TBI. The three most used medication groups in this population were tranquilizers/sleep medications, antidepressants, and opioids.[18] However, the specific contribution of antidepressants—and in particular their serotonergic profile—to outcomes following TBI has not been specifically investigated.

It is becoming increasingly important to understand whether serotonergic antidepressants affect short-term outcomes in TBI, given their prevalence in the aging population at high risk of falls. In this study, we used a nationwide cohort to examine the association between preinjury antidepressant use, their serotonergic profile, and short-term outcomes after TBI. We hypothesized that serotonergic antidepressant use is not independently associated with higher short-term mortality or the need for ANOs but that serotonergic profile (e.g., weak vs strong serotonin reuptake inhibition) may modulate the risks.

Methods

Patients

We retrospectively identified all ward admissions due to TBI (primary diagnosis ICD-10 S06.*) in patients aged 16 years or older treated in Finnish hospitals and healthcare wards between January 1, 2005, and December 31, 2018, using data from the Care Register for Health Care.[15,17] The database, which is maintained by the National Institute for Health and Welfare (Helsinki, Finland) under a legal obligation, includes every discharge from a hospital ward in Finland and includes data on surgical operations.[19] Transfers between and within healthcare providers related to a particular admission episode were combined as one admission. The study included data from 339 hospitals or other health care units treating TBI, 5 of which provide neurosurgery services. The first admission of each patient during the study period was included. We excluded patients with chronic subdural hematoma (ICD-10 codes S06.5 or I62.0) who had undergone surgical evacuation of the hematoma (operation codes AAD10 or AAD12). In addition, patients without follow-up data (n = 498) were excluded. The primary outcome was all-cause mortality within 30 days of TBI admission, while secondary outcomes included ANO and length of hospital stay.

The accuracy of ICD-10 TBI coding in the underlying study cohort has been earlier assessed by reviewing medical records of randomly selected patients who met the inclusion criteria (primary discharge diagnosis S06.*) at Turku University Hospital (Turku, Finland) and North Karelia Central Hospital (Joensuu, Finland).[15] Of the 300 reviewed cases, 294 met the diagnostic criteria for S06.*, leading to a positive predictive value of 0.98.[15]

Definitions

Pharmacy purchases of antidepressants within 90 days before TBI were recognized using Anatomical Therapeutic Chemical (ATC) codes (eTable 1). Antidepressants are only available on prescription from pharmacies in Finland.[20] The daily pill counting method based on the standardized defined daily dose (eTable 1) was used to estimate the usage of prescribed medication on the day of TBI. Antidepressant therapy was classified by serotonergic profile (strong, intermediate, weak) based on serotonin transporter inhibition (eTable 1).[21-25]

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Used antidepressant was determined to be the last purchased before TBI. Because anticoagulation with VKAs is associated with poorer short-term outcome of TBI, the usage of VKA was determined using the same methodology, as previously described.[15] The usage of DOAC was also determined, as previously described.[15] Patients who had purchased antidepressants or VKAs within 90 days before TBI admission, but for whom the duration of medication purchased did not cover the day of TBI, were excluded (Figure 1). Patients with purchases of parenteral anticoagulants without concomitant purchases of VKAs or DOACs were also excluded. Skull and facial fractures were defined by ICD-10 code S02.* as a secondary diagnosis. Comorbidities were identified using data from the included registries.[26] ANOs were identified, as previously defined.[27] Length of stay was calculated to include both ward care and interhospital transfers.

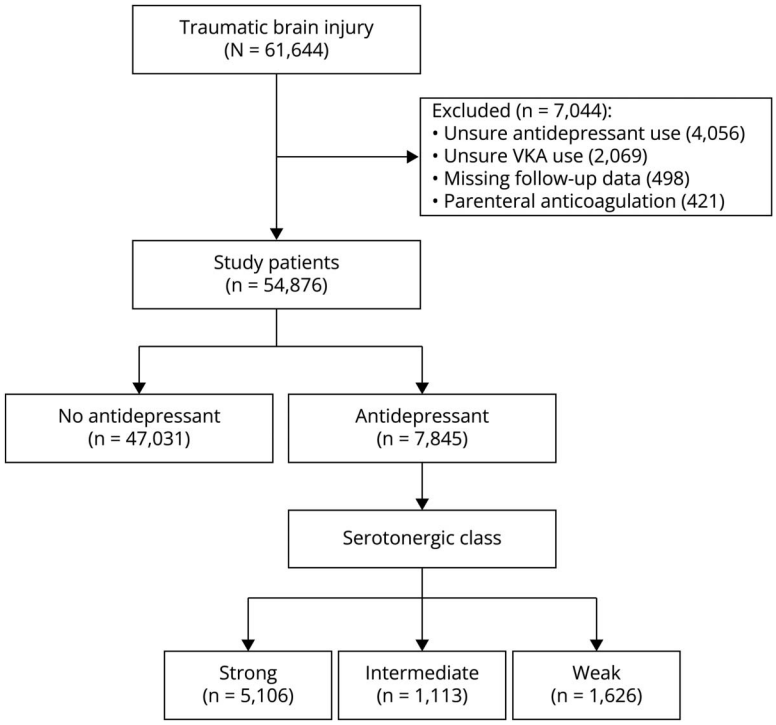
Standard Protocol Approvals, Registrations, and Patient Consents

Data from the Finnish Care Registry of Health Care Registry, the Finnish Cancer Registry and prescription medication purchase records (covering ATC codes, dosage strength, quantity and date of purchase) and drug reimbursement permissions, were obtained from the National Institute for Health and Welfare of Finland/Findata (permission no. THL/164/14.02.00/2021). Mortality data were obtained from the nationwide Cause of Death Registry maintained by Statistics Finland (permission no. TK-53-484-20). The collection and reporting of data in these registers is required by law and ensures complete coverage of the Finnish population. Follow-up data were available for all patients. As this was a retrospective, nationwide registry study, individual hospital review board approval and informed consent were waived by law, and patients were not contacted. The data used in this study were provided by the Finnish Social and Health Data Permit Authority Findata under license and can be obtained from Findata (findata.fi) on permission. The legal basis for the processing of personal data is public interest and scientific research in accordance with the EU General Data Protection Regulation 2016/679 (Article 6(1)(e) and Article 9(2)(j)) and the Finnish Data Protection Act (Sections 4 and 6).

Statistical Methods

Baseline features were analyzed with χ^2 test or analysis of variance as appropriate. Mortality was analyzed with the Kaplan-Meier method, log-rank test, and adjusted Cox-regression. Proportional hazard assumptions were examined with Schoenfeld residuals. ANOs were analyzed with modified Poisson regression.[28] Admission duration was analyzed with linear regression (log-transformed and standardized dependent variable). Regression analyses were adjusted with age, sex, and comorbidities, as listed in Table 1 (except for atrial fibrillation), skull or facial fracture, VKA treatment, initial admission location, and stratified by the year of TBI admission. Potential modulative influence of combined

Figure 1 Study Flowchart



VKA = vitamin K antagonist.

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antidepressant and VKA or DOAC therapies was studied with interaction analysis in adjusted regression models. The results are given as the mean, median, percentage, hazard ratio (HR), or relative risk (RR) with a 95% CI, interquartile range (IQR), or ±SD. Statistical significance was inferred at $p < 0.05$. Analyses were performed with SAS version 9.4 (SAS Institute, Inc., Cary, NC).

Data Availability

By law, we are not permitted to disclose data to third parties. Requests to access the data set may be sent to Findata, Helsinki, Finland.

Results

The total study population included 54,876 patients with TBI, of whom 7,845 (14.3%) were classified as antidepressant users at the time of TBI. Patients with antidepressant were older, more often female, and had higher comorbidity burden compared with patients without antidepressant (Table 1). Initial admission location was more frequently local health center than hospital with surgical services in patients with antidepressant treatment. The serotonergic class of used antidepressant was strong in 65.1%, intermediate in 14.2%, and weak in 20.7% of the patients (Table 2). Weak serotonergic

Table 1 Baseline Features of Patients With Traumatic Brain Injury With and Without Antidepressant

Variable	Antidepressant (N = 7,845)	No antidepressant (N = 47,031)	*p* Value
Age, y, mean (SD)	64.7 (20.0)	60.6 (22.6)	<0.0001
Female, %	56.0	41.4	<0.0001
Comorbidities, %			
Alcohol abuse	23.1	13.7	<0.0001
Atrial fibrillation	15.7	13.7	<0.0001
Cerebrovascular disease	22.8	17.2	<0.0001
Chronic pulmonary disease	15.6	11.0	<0.0001
Coagulopathy	0.7	0.8	0.450
Dementia	17.1	9.1	<0.0001
Diabetes	19.2	13.1	<0.0001
Drug abuse	5.3	1.5	<0.0001
Heart failure	14.5	10.6	<0.0001
Heart valve disease	4.2	4.0	0.312
Hypertension	41.5	31.7	<0.0001
Liver disease	4.0	2.5	<0.0001
Malignancy	12.7	10.6	<0.0001
Myocardial infarction	7.2	6.0	<0.0001
Paralysis	1.2	0.7	<0.0001
Peripheral vascular disease	4.3	3.6	0.003
Psychotic disorder	14.8	4.1	<0.0001
Rheumatic disease	5.2	4.4	0.001
Renal failure	1.9	1.8	0.250
Skull or facial fracture, %	6.5	9.0	<0.0001
Vitamin K antagonist, %	8.0	6.7	<0.0001
Direct oral anticoagulant, %	0.9	0.9	0.885
Admission location, %			<0.0001
University hospital	28.3	32.8	
Other hospital	38.9	44.1	
Health center	32.8	23.1	

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class was associated with older age and higher somatic comorbidity burden. VKA was used more frequently by patients with antidepressant (Table 1). Weak serotonergic class was associated with higher VKA use (Table 2). Patients using antidepressant had lower frequency of skull or facial fractures, with the lowest frequency in weak serotonergic class of antidepressant (Tables 1 and 2). The most common antidepressant type was SSRI (59.8%), followed by atypical (19.9%), SNRI (15.3%), and tricyclic antidepressant (5.0%). The most common antidepressant was citalopram (27.4%) (eTable 1). Patients with SSRI, SNRI, and tricyclic antidepressants were younger and overall had less comorbidities than those with atypical antidepressants (eTable 2).

Patient characteristics varied depending on the place of admission. Patients admitted directly to university hospitals

Table 2 Baseline Features of Traumatic Brain Injury Patients With Strong, Intermediate, or Weak Serotonergic Antidepressant and Patients Without Antidepressant

	Antidepressant serotonergic class				Between group *p* value	
Variable	**Strong (N = 5,106)**	**Intermediate (N = 1,113)**	**Weak (N = 1,626)**	**None (N = 47,031)**	**All**	**Within antidepressants**
Age, y, mean (SD)	64.3 (20.4)	59.7 (19.5)	69.2 (18.1)	60.6 (22.6)	<0.0001	<0.0001
Female, %	58.3	54.0	49.9	41.4	<0.0001	<0.0001
Comorbidities, %						
Alcohol abuse	21.1	27.2	26.5	13.7	<0.0001	<0.0001
Atrial fibrillation	15.3	11.2	20.1	13.7	<0.0001	<0.0001
Cerebrovascular disease	22.4	19.2	26.5	17.2	<0.0001	<0.0001
Chronic pulmonary disease	15.1	17.6	16.1	11.0	<0.0001	0.089
Coagulopathy	0.7	0.5	0.8	0.8	0.639	0.541
Dementia	17.1	10.7	21.4	9.1	<0.0001	<0.0001
Diabetes	18.6	21.5	19.3	13.1	<0.0001	0.088
Drug abuse	4.6	7.8	5.6	1.5	<0.0001	0.0002
Heart failure	13.9	11.4	18.7	10.6	<0.0001	<0.0001
Heart valve disease	4.0	3.0	5.5	4.0	0.004	<0.0001
Hypertension	40.5	38.3	47.0	31.7	<0.0001	<0.0001
Liver disease	3.5	4.9	5.1	2.5	<0.0001	0.004
Malignancy	12.7	10.4	14.2	10.6	<0.0001	0.014
Myocardial infarction	7.1	5.0	9.3	6.0	<0.0001	<0.0001
Paralysis	1.3	1.4	1.0	0.7	<0.0001	0.529
Peripheral vascular disease	4.1	4.0	4.9	3.6	0.011	0.389
Psychotic disorder	13.1	20.1	16.4	4.1	<0.0001	<0.0001
Rheumatic disease	5.0	5.8	5.8	4.4	0.004	0.305
Renal failure	1.8	2.0	2.5	1.8	0.190	0.205
Skull or facial fracture, %	6.5	8.3	5.6	9.0	<0.0001	0.020
Vitamin K antagonist, %	7.7	6.2	10.3	6.7	<0.0001	0.0002
Direct oral anticoagulant, %	0.7	0.9	1.4	0.9	0.127	0.061
Admission location, %					<0.0001	<0.0001
University hospital	27.5	33.6	27.4	32.8		
Other hospital	39.8	39.7	35.7	44.1		
Health center	32.8	26.7	36.8	23.1		

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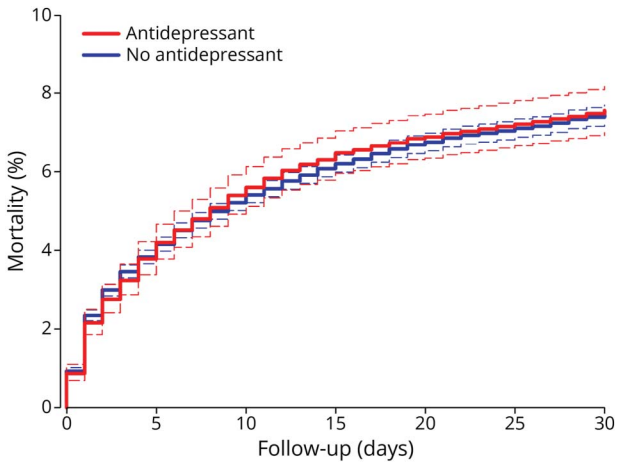
were younger, more often male, and had fewer comorbidities, whereas patients admitted to other hospitals or health centers were older, more often female, and had a higher burden of comorbidities, especially cardiovascular disease. Antidepressant use was also more common in patients initially treated outside university hospitals (eTable 3). The distribution of primary ICD-10 TBI diagnoses is presented in eTable 4.

Mortality

Of all TBI patients, 4,105 died within 30 days. The 30-day mortality was 7.6% in patients with antidepressant and 7.5% in patients without antidepressant (Figure 2). Mortality did not differ by antidepressant use in nonadjusted (p = 0.746) or multivariable adjusted analysis (adjusted HR 0.98; 95% CI 0.90–1.07 for antidepressant vs no antidepressant; p = 0.696). Mortality was 7.1% in strong, 6.3% in intermediate, and 10.0% in weak serotonergic group. In nonadjusted analysis, the mortality rate was higher in weak serotonergic group compared with control patients without antidepressant (p = 0.0002) but did not differ between strong or intermediate serotonergic groups and control patients. In multivariable analysis adjusted for age, sex, comorbidities, skull or facial fracture, VKA use, study year, and treatment location, there was no difference in mortality by serotonergic class of antidepressant (p = 0.312).

Adjusted HR of 30-day death was 0.94 (95% CI 0.84–1.05; p = 0.262) for strong serotonergic group vs controls, 0.95 (95% CI 0.75–1.20; p = 0.646) for intermediate serotonergic group vs controls, and 1.12 (95% CI 0.95–1.31; p = 0.177) for weak serotonergic group vs controls. There was no difference in mortality between strong vs weak (adjusted HR 0.84; 95% CI 0.70–1.01; p = 0.068), strong vs intermediate (adjusted HR 0.99; 95% CI 0.77–1.28; p = 0.948), or intermediate vs weak (adjusted HR 0.85; 95% CI 0.64–1.12; p = 0.245) serotonergic antidepressant users. Antidepressant type was not associated with mortality (adjusted p = 0.352). There was no interaction between VKA and antidepressant use (interaction p = 0.849), VKA and serotonergic class (interaction p = 0.927), or VKA and antidepressant type (interaction p = 0.478) in 30-day mortality. Patients with VKA had higher mortality (14.3% vs 6.3%; adjusted HR 1.38; 95% CI 1.26–1.52; p < 0.0001). There was no interaction between DOAC and antidepressant use, class, or type regarding mortality (eTable 5).

Figure 2 30-Day Mortality After Traumatic Brain Injury by the Usage of Antidepressants



Dashed lines represent 95% CIs.

Acute Neurosurgical Operations

ANO was performed to 4,586 patients with TBI. The ANO rate was 6.8% in the antidepressant users' group and 8.6% in controls without antidepressant (nonadjusted p < 0.0001). Patients with antidepressant had lower likelihood for ANO than patients without antidepressant (adjusted RR 0.89; 95% CI 0.82–0.97; p = 0.007). The ANO rate was 6.5% in strong, 7.6% in intermediate, and 7.2% in weak serotonergic group.

Serotonergic class of antidepressant was associated with likelihood of ANO (adjusted p = 0.043). RR of ANO was lower in strong serotonergic group vs controls (adjusted RR 0.88; 95% CI 0.79–0.97; p = 0.015) but similar in intermediate (adjusted RR 0.89; 95% CI 0.73–1.09; p = 0.272) and weak (adjusted RR 0.92; 0.78–1.09; p = 0.331) serotonergic groups and control groups. Antidepressant type was not associated with ANO (adjusted p = 0.069). There was no interaction between VKA and antidepressant use (interaction p = 0.143), VKA and serotonergic class (interaction p = 0.211), or VKA and antidepressant type (interaction p = 0.084) regarding likelihood of ANO. Patients with VKA had higher frequency of ANO (9.4% vs 8.3%; adjusted RR 1.27; 95% CI 1.14–1.41; p < 0.0001). There was no interaction between DOAC and antidepressant use regarding ANOs (eTable 5).

Admission Duration

The median length of all TBI admissions was 3 days (IQR 2–10 days). The duration was 4 days (IQR 2–11) in the antidepressant users' group and 3 days (IQR 2–10) in the control group (nonadjusted p < 0.0001). There was no difference in admission length by antidepressant use in multivariable adjusted analysis (p = 0.504). The median admission duration was 4 days (IQR 2–11) in strong, 3 days (IQR 2–9), in intermediate, and 4 days (IQR 2–12) in weak serotonergic group (nonadjusted p < 0.0001). Serotonergic class or antidepressant type was not associated with admission duration in multivariable-adjusted analysis (p = 0.072 and p = 0.103, respectively). The median admission duration was 5 days (IQR 2–15) in patients with VKA vs 3 days (IQR 2–10) in patients without VKA. There was no interaction between VKA and antidepressant use (interaction p = 0.143), VKA and serotonergic class (interaction p = 0.281), or VKA and antidepressant type (interaction p = 0.104) in admission duration. Treatment with VKA was associated with longer admission in the nonadjusted analysis (p < 0.0001), but not after multivariable adjustments (p = 0.564). There was no interaction

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between DOAC and antidepressant use, class, or type regarding admission duration (eTable 5).

Discussion

This large nationwide, population-based study found that preinjury serotonergic antidepressant use was not associated with increased 30-day mortality, increased need for ANOs, or prolonged hospitalization after TBI. Serotonergic class or type of antidepressant was not associated with any of the outcomes. In addition, antidepressant use or serotonergic class or type of antidepressant did not to have interaction with increased mortality among VKA users. These findings are reassuring given high incidence of head injury and the widespread use of serotonergic antidepressants in older adults, as well as previous reports of serotonergic antidepressant-related bleeding complications.

This study addresses an issue that is a daily concern for many clinicians working in a variety of clinical settings. Yet, there is a paucity of research data on this topic. Our current findings are consistent with the limited previous research base that has examined the safety of serotonergic antidepressants in the context of traumatic intracranial hemorrhage. In, a single-center study setting, Isokuortti et al. found no association between serotonergic antidepressant use and traumatic intracranial hemorrhage in either prospective or retrospective cohorts of patients with mild TBI, regardless of antidepressant type.[13,14] These results are in line with earlier evidence that questioned the clinical relevance of SSRI-associated bleeding risk.[12,29]

Patients taking antidepressants were older, more often female, and had more comorbidities than patients not taking antidepressants. These characteristics may be related to a higher prevalence of low energy trauma mechanisms (e.g., falls) and diffuse injury patterns that are less likely to result in focal mass lesions that can be surgically removed. This is supported by the fact that skull or facial fractures—indicative of high-energy impacts—were less common in antidepressant users and least common in the low serotonergic group. In addition, a greater proportion of patients in the antidepressant group was admitted to health centers rather than university hospitals, suggesting a possible difference in the severity classification of care. Overall, these results suggest that the lower ANO rate may not be directly attributable to the pharmacologic effects of antidepressants, but rather to a different clinical and injury profile in these patients, including milder or in some cases more diffuse injuries that do not meet surgical thresholds. These findings are in line with our previous studies on the Finnish epidemiology of TBI.[27,30] We report the distribution of primary ICD-10 diagnoses. Although concomitant intracranial lesions not captured by ICD-10 codes could contribute to outcome differences, our adjusted models accounted for key demographic, clinical, and treatment-related factors, and there is no clear rationale for systematically different injury severity or mechanisms based on antidepressant use alone.

The proposed mechanism behind platelet dysfunction in the primary hemostasis is related to decreased platelet serotonin levels due to SSRI-mediated inhibition of the serotonin transporter.[31] SSRIs and other serotonin reuptake inhibiting antidepressants reduce platelet serotonin levels, which impairs platelet aggregation and thus increases bleeding tendency.[32-34] This pharmacodynamic property is theoretically important in the context of TBI, where diffuse microvascular injury is common.

Previously, Verdel et al.[35] found in a retrospective case-control study that the current use of serotonergic antidepressants was associated with a modest but statistically significant increased risk of intracranial hemorrhage, although they did not distinguish between traumatic and nontraumatic causes of hemorrhage. The increased risk was most pronounced in new users. However, no clear association was found between affinity for serotonin transporters or 5-hydroxytryptamine receptor 2A affinity and bleeding risk. However, a systematic review article reported that serotonergic antidepressants may increase the risk of nontraumatic intracranial hemorrhage. It was found that the risk of hemorrhage may be increased with strong inhibitors of serotonin reuptake compared with weak inhibitors. In the studies analyzed, 4 of 10 studies showed an increased risk of intracranial hemorrhage associated with SSRIs, with the 2 largest studies indicating a moderate effect. Three of 5 studies showed an increased risk of intracranial hemorrhage associated with strong serotonin reuptake inhibitors.[36] However, in this study, stratification by serotonergic profile did not exhibit a dose-response relationship or interaction with TBI outcomes. This lack of association in our study may be because traumatic rather than spontaneous bleeding occurred in the cohort, in which the risk of bleeding is determined more by the mechanism of injury than by serotonergic pharmacodynamics.

As shown previously,[15] preinjury VKA use was associated with increased 30-day mortality and need for ANOs, consistent with findings of meta-analysis on VKA-related intracranial hemorrhage risk in TBI.[37] We found no interaction between VKA use and antidepressant use, serotonergic class, or type of antidepressant, suggesting that these exposures independently affect early outcomes. Furthermore, we found no interaction between antidepressant use, class, or type and DOAC use on the studied outcomes. These findings emphasize the importance of early identification and reversal of anticoagulation in patients with TBI, regardless of concomitant serotonergic antidepressant medication use. The impact of adenosine diphosphate inhibitors on the current outcomes was not investigated as we have previously reported a lack of association in another Finnish population-based study.[15] Chronic subdural hematomas were excluded from this study because their pathogenesis and clinical course differ substantially from acute traumatic intracranial hemorrhage. It is

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known that symptoms typically develop with a delay of up to 2 months after trauma,[38] and many patients with chronic subdural hematoma have no recognizable previous trauma at all.[39]

Our study has several strengths, including a large sample size, broad linkage to national registries, and the use of standardized classifications for medication exposure. The study includes all Finnish hospitals involved in the treatment of TBI and all tertiary care centers providing neurosurgical services. The use of several national registries—all of which are required by law to ensure systematic data collection and reporting—provides a representative overview of the entire Finnish population. However, some limitations must be acknowledged. Although the multivariable models were adjusted for key baseline characteristics, the observed lower likelihood of ANOs among antidepressant users—particularly those receiving strong serotonergic agents—may still be due to residual confounding related to mechanism of injury or frailty that is not fully captured by the available covariates. We were not able to assess medication adherence, changes in antidepressant use after injury or the severity of underlying psychiatric conditions. The risk of bleeding may be dose-dependent, at least in the obstetric context.[39] In this study, we were unable to investigate the relationship between the different drug doses and outcomes. In addition, we can only draw limited conclusions about patient outcomes due to a lack of long-term follow-up data. A limitation of this study is that the national registry data do not contain variables for typical classifications of injury severity, such as Glasgow Coma Scale score on admission, postinjury amnesia and duration of unconsciousness, or radiologic scoring systems. Therefore, we report the distribution of primary ICD-10 TBI diagnoses as descriptive context. However, we consider 30-day mortality to be a robust indicator of severe complications. In addition, the need for ANOs in the population reflects the rate of intracranial space-occupying hemorrhagic lesions and the length of hospital stay reflects the severity of TBI and the need for closer clinical monitoring on the wards. We conclude that this 3-dimensional outcome assessment provides a solid overview that preinjury antidepressant use does not worsen early clinical outcomes after TBI. Finally, care resources and processes differ by admission site, as university hospitals provide neurosurgical and neurocritical care services that central and regional hospitals and health centers do not. Such differences in organization, monitoring, and acute interventions could influence outcomes, but process-level data were not available in the national registries. It is also possible that comorbidities and frailty influenced initial triage decisions and contributed to the higher proportion of antidepressant users first admitted to non-neurosurgical hospitals. However, care in Finland is organized so that all patients with a prognosis of recovery after TBI are referred from health centers and other hospitals to a university hospital if their condition is more severe than a mild, uncomplicated brain injury and neurosurgical and neurocritical care services are required.

We found that preinjury serotonergic antidepressant use was not associated with increased 30-day mortality, need for ANOs, or prolonged hospitalization after TBI. Serotonergic class or type of antidepressant was not adversely associated with any of the outcomes. No interaction was found between VKA use and antidepressant use, serotonergic class, or type of antidepressant, indicating that antidepressants do not exacerbate the adverse impact of VKA on early outcomes. These findings suggest that preinjury antidepressant use does not worsen early clinical outcomes after TBI.

Author Contributions

J.P. Posti: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. A. Tornio: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.O. Ruuskanen: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. V. Kytö: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

Study Funding

State Research Funding (VTR) (Finland) and grant funding of the Paulo Foundation, the Paavo Nurmi Foundation, and the Finnish Foundation for Cardiovascular Research. The funding sources had no role in study design; in the collection, analysis, and interpretation of data; in the writing of the report; and in the decision to submit the article for publication.

Disclosure

J.P. Posti has received speaker's fees from Sanofi S.A., the Finnish Medical Association, Wellbeing services county of North Karelia, and Finnish Association of Otorhinolaryngology - Head and Neck Surgery; received travel expenses reimbursement and expert fee from the National Institute of Neurological Disorders and Stroke; and is supported by the Research Council of Finland and the Sigrid Jusélius Foundation. A. Tornio has received scientific consultancy fees from Aplagon Oy and Clinical Research Services Turku-CRST Oy, and speaker's fees from Bayer, Farmasian oppimiskeskus and Pfizer. J.O. Ruuskanen has received a scientific consultancy fee (Sandoz), speaker fee (Merck), travel/congress sponsorship (Bristol Myers Squibb and Bayer), and holds shares (Medbase Ltd.). V. Kytö reports no relevant disclosures. Go to Neurology.org/N for full disclosures.

Publication History

Received by *Neurology*® June 24, 2025. Accepted in final form November 12, 2025. Submitted and externally peer reviewed. The handling editor was Associate Editor Rebecca Burch, MD.

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